

Varun Khajuria | PhD Researcher

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Profile

PhD candidate on mountain permafrost hydrology and rock–glacier dynamics in the Southwestern Alps; working in InSAR/optical RS, GEE/Python geospatial analytics, and cryosphere modelling for water resources.

Education

University of Pisa

PhD, Geosciences & Environment

Italy

2023–present

Thesis: Permafrost ice in the SW Alps: a long-term water reservoir; *Advisors:* A. Ribolini, S. Singh

Punjab Engineering College

M.Tech, Water Resources

Chandigarh

2021

Distinction; *Thesis:* Snowmelt runoff of a catchment

Bharati Vidyapeeth Deemed Univ.

B.Tech, Civil Engineering

2017

First class with Distinction

Professional Experience

School of Geosciences, University of Aberdeen

Visiting PhD Researcher

Scotland, UK

2024

Permafrost field methods; satellite data integration for RG thickness/kinematics

Water Resources Dept., IIRS (ISRO)

Research Scientist

Dehradun, India

2021–2023

Glacier dynamics (SAR/optical), snowmelt runoff modelling, hydrological simulations; papers and national/international talks

Kapahi Contracts Pvt. Ltd.

Civil Engineer

India

2017–2019

Site supervision, QA/QC, survey workflows

Selected Publications

1: Ribolini, A., Forte, E., Khajuria, V., Colucci, R. R., Paro, L., Guglielmin, M. (2025). Scratching beneath the surface: Using Ground-Penetrating Radar to disentangle pronival ramparts, embryonic rock glaciers and moraines (Gardetta Plateau, Southwestern Alps). *Geomorphology*, 474, 109647. <https://doi.org/10.1016/j.geomorph.2025.109647>

2: Kumar, M., Khajuria, V. et al. Process-based models for Himalayan hydrology. *Earth Syst. Environ.*

3: Khajuria, V.* et al. Snowmelt runoff estimation Using Combined Terra-Aqua MODIS Improved Snow product in Western Himalayan River Basin via degree day modelling approach. *Environmental Challenges* 8:100585

Grants & Awards

2024: Erasmus+ KA131 Extra-EU Mobility Grant (Univ. of Aberdeen)

2019–2021: MHRD–AICTE GATE Scholarship | 2013–2017 PMSSS Scholarship

Selected Conferences

2025: SGI-SIMP (Padova, Italy): Rock Glacier detecting in the SW Apl

2023: NCPOR (Goa, India): Ice thickness & volume modelling, Zaskar Basin

Skills

RS: Sentinel-1 InSAR; S2/Landsat; PlanetScope | Geospatial: GEE, QGIS/ArcGIS, DEMs | Modelling: snow/ice melt, RG thickness/kinematics | Programming: Python (NumPy, GeoPandas, Rasterio), R |

Service & Leadership

Associate Editor, *Earth Science Informatics* (Springer) | Life Member, ISRS; former Affiliate, ASCE

References

Prof. Adriano Ribolini; Dr. Praveen K. Thakur; Dr. B.R. Nikam — *Available upon request*